

Stochastic Partial Differential Equations: Some Applications and Theoretical Results

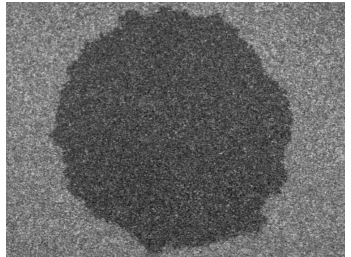
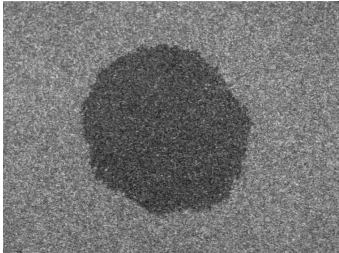
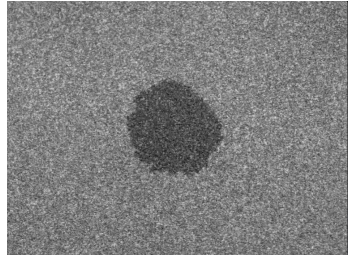
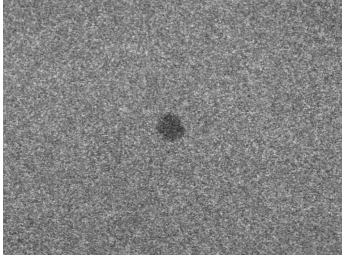
Nicholas Eisenberg

May 19, 2022

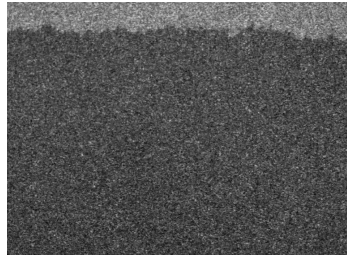
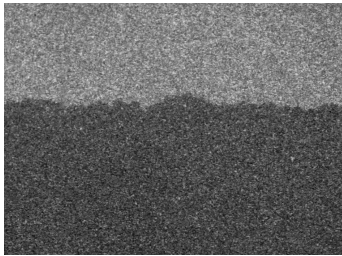
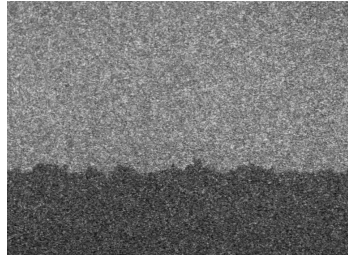
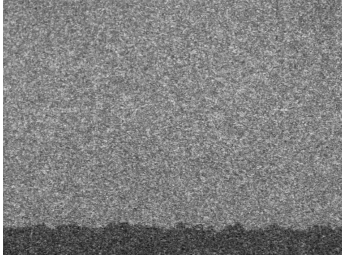
Some of the Questions

- We want to model the growth of a cluster of cells undergoing mitosis. How can we include the randomness in our model? (KPZ- equation)
- We are investors looking for arbitrage opportunities on call options. Can we calculate a fair price? (Black-Scholes-Merton model)
- We want to model the evolution of heat of a piece of an initially heated wire. How can we account for random outside sources of heat that affect the wires temperature? (SHE)

Macroscopic Cell Growth



Microscopic Cell Growth



Solving the KPZ Equation

Sir Martin Hairer
KBE FRS



Martin Hairer at the Royal Society admissions day in London, July 2014

Born	14 November 1975 (age 45) Geneva, Switzerland
Citizenship	Austrian British
Education	College Claparede, Geneva
Alma mater	University of Geneva
Spouse(s)	Xue-Mei Li (m. 2003) ^{[1][2]}
Awards	Whitehead Prize (2008) Philip Leverhulme Prize (2008) Wolfson Research Merit Award (2009) Fermat Prize (2013) Fröhlich Prize (2014) Fields Medal (2014) Breakthrough Prize in Mathematics (2021)
	Scientific career
Fields	Probability theory ^[3] Analysis ^[3]
Institutions	Imperial College London University of Warwick New York University ^[1]
Thesis	Comportement Asymptotique d'Équations à Dérivées Partielles Stochastiques  (2001)
Doctoral advisor	Jean-Pierre Eckmann ^[4]
Website	hairer.org 

- ▶ $\frac{\partial}{\partial t} u(t, x) = \frac{1}{2} \Delta_x u(t, x) + \frac{\lambda}{2} (\nabla_x u(t, x))^2 + \dot{W}(t, x)$
- ▶ M. Hairer. Solving the KPZ equation. *Annals of Mathematics*, vol. 178, 2013.
- ▶ M. Hairer. A theory of regularity structures. *Inventiones mathematicae*, vol. 198, 2014.

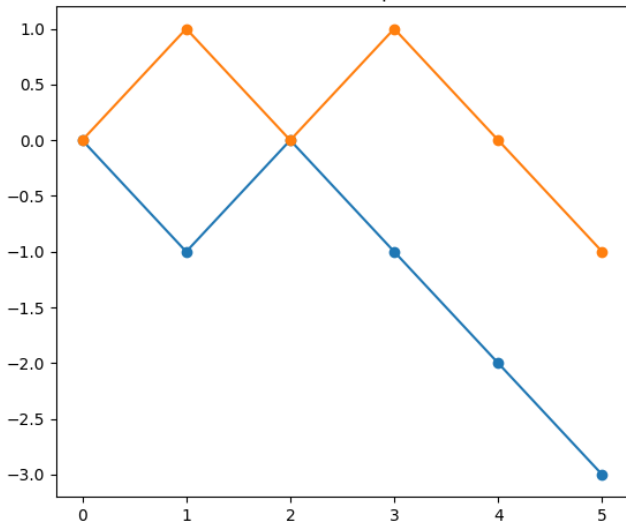
Fields Medal in 2014

Noise

- Essentially, the randomness is accounted for by a random noise, usually Gaussian.
- The noise can be thought of as the derivative of a non-differentiable function. In the the case of a SODE, the derivative of Brownian motion and in the case of a SPDE, the derivative of a Brownian sheet.
- There are more general noises, but it is still helpful to keep the above "rough" idea in our heads.

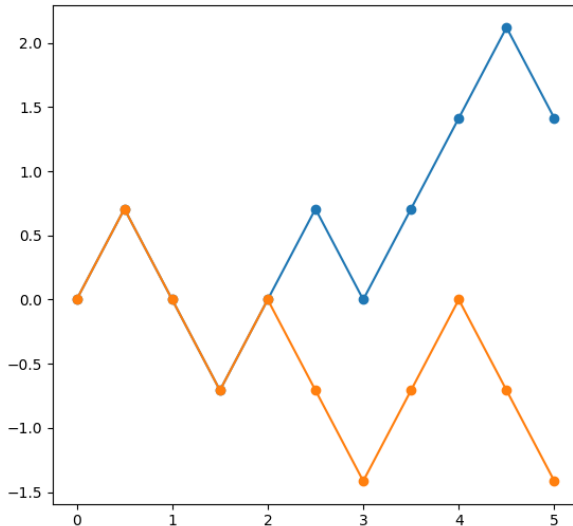
Random Walk

2 sample paths of a random walk
jump size = 1.0 and step size = 1.0
number of steps = 5



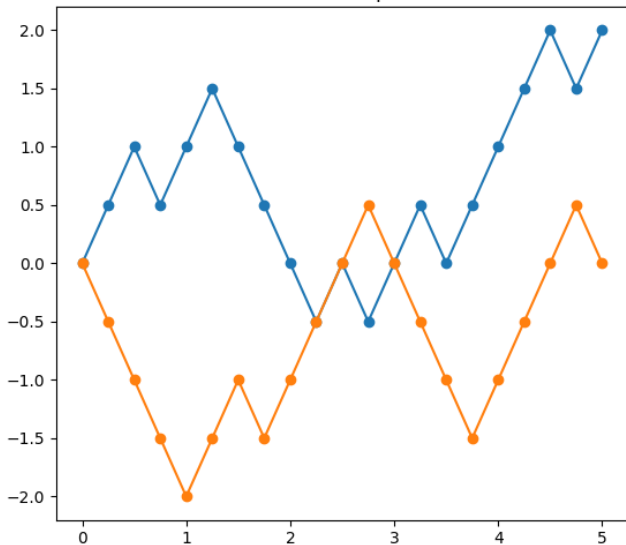
Random Walk

2 sample paths of a random walk
jump size = 0.707 and step size = 0.5
number of steps = 10

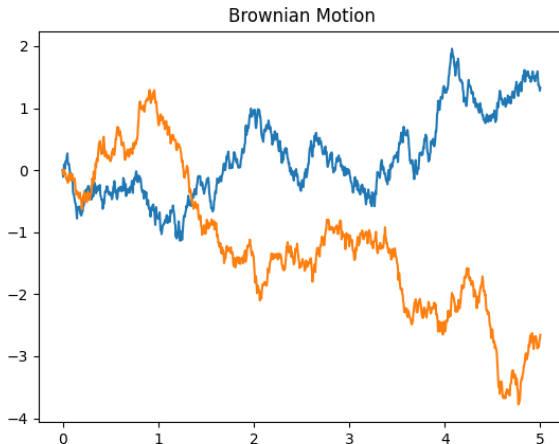


Random Walk

2 sample paths of a random walk
jump size = 0.5 and step size = 0.25
number of steps = 20

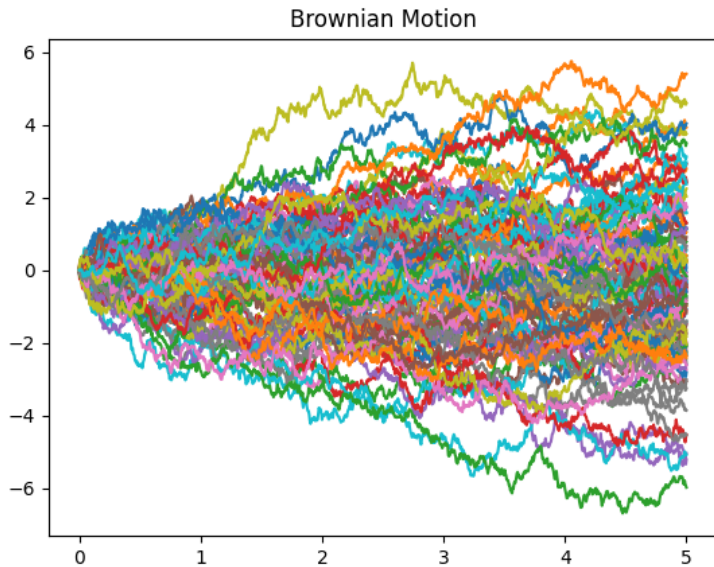


Random Walk $\rightarrow$ Brownian Motion



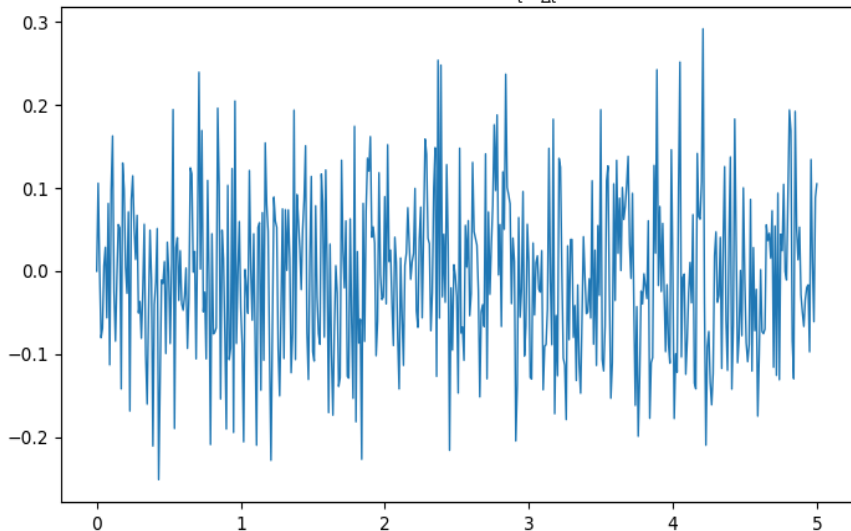
- Property of Brownian Motion: $B_t \sim B_{t-\Delta t} + \text{Normal}(0, \Delta t)$.
- Recall that $\text{Normal}(0, \Delta t) = \sqrt{\Delta t} \text{Normal}(0, 1)$

Square Root Growth



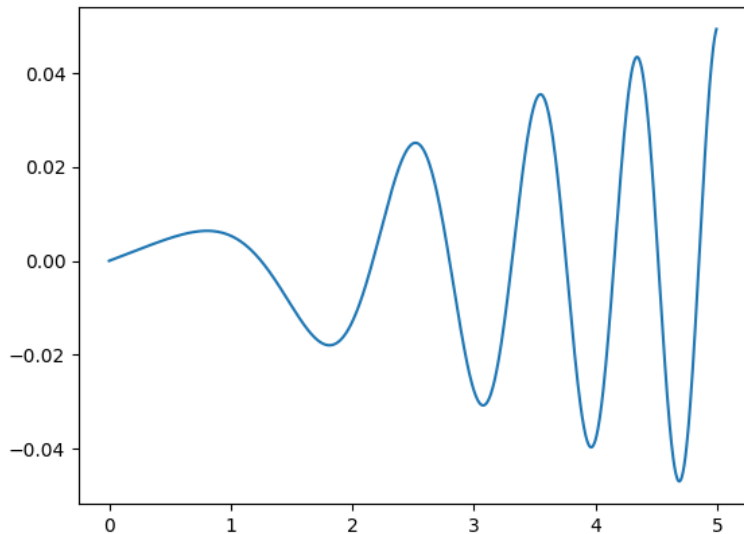
White Noise

White Noise differential: $dB_t \approx \int_{t-\Delta t}^t dB_s = B_t - B_{t-\Delta t}$:



Increments of a Smooth Function for Reference

Increments, $df(x_i) = f(x_i) - f(x_{i-1})$, for $f(x) = \sin(x^2)$



What Can We Do with Brownian Motion?

- S_0 = stock price at entry S_t = stock price after t time-periods

$$S_t = S_0(1 + R_t), \quad R_t := \frac{S_t - S_0}{S_0} \sim \text{Normal}(\mu t, \sigma^2 t), \quad \mu \in \mathbb{R}, \sigma \in (0, \infty).$$

- We can model this with a stochastic differential equation.

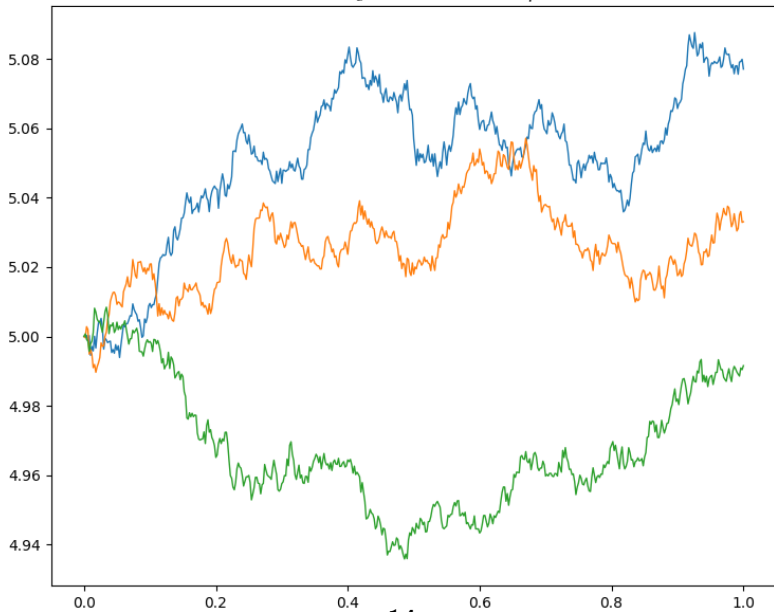
$$R_t = \frac{dS_t}{S_t} = \mu dt + \sigma dB_t \quad \text{or} \quad dS_t = \mu S_t dt + \sigma S_t dB_t$$

- It can be shown that $\mu dt + \sigma dB_t \sim \text{Normal}(\mu \cdot \Delta t, \sigma^2 \cdot \Delta t)$
- The legitimate representation of the stochastic ODE is the following:

$$S_t(\omega) = S_0 + \int_0^t \mu S_k(\omega) dk + \int_0^t \sigma S_k(\omega) dB_k(\omega).$$

Geometric Brownian Motion: $dS_t = \sigma S_t dB_t + \mu S_t dt$

$(a, b) = (0, 1)$, $X_a = 5$, $\sigma = 0.01$, and $\mu = 0.001$



Black-Scholes-Merton Option Model

Fischer Black



Myron Scholes



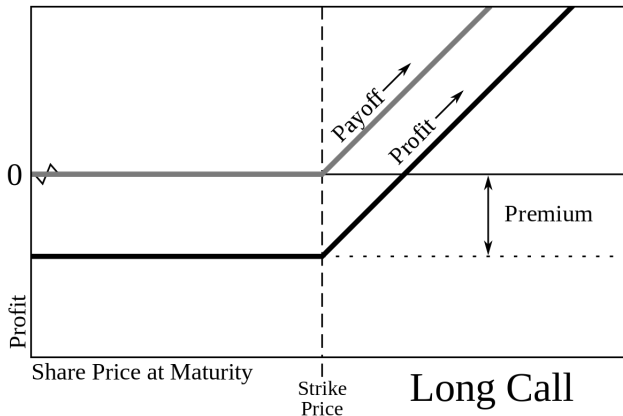
Robert C. Merton



- Scholes and Merton received the Nobel Memorial Prize in Economics - 1997.
- Unfortunately Black passed away due to cancer before the award was presented.

Fair Price for a European Call Option

- Consider a contract that permits the holder to purchase, at precisely time $t = T$, 1 share of a stock at price K regardless of the price of the stock at time $t = T$.
- What is the fair price for this contract at time $t = 0$?



It can be shown that $f(T, S_0)$ is the fair price where f solves the following partial differential equation...

$$\begin{cases} \frac{\partial f}{\partial s}(s, x) = \frac{1}{2}\sigma^2 x^2 \frac{\partial^2 f}{\partial x^2}(s, x) + rx \frac{\partial f}{\partial x}(s, x) - rf(s, x) & (s, x) \in [0, T] \times \mathbb{R}_+, \\ f(0, x) = (x - K)^+ & x \in (0, \infty). \end{cases}$$

The solution is given by

$$f(t, x) = x\Phi(g(t, x)) - Ke^{-rt}\Phi(h(t, x)),$$

where

$$g(t, x) = \left[\ln\left(\frac{x}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)t \right] / (\sigma\sqrt{t}),$$

$$h(t, x) = g(t, x) - \sigma/\sqrt{t},$$

$$\Phi(t) = \frac{1}{\sqrt{2\pi}} \int_0^t \exp\left(-\frac{x^2}{2}\right) dx.$$

Stochastic Heat Equation

$$\begin{cases} \frac{\partial}{\partial t} u(t, x) - \frac{1}{2} \frac{\partial^2}{\partial x^2} u(t, x) = \lambda u(t, x) \frac{\partial^2}{\partial t \partial x} W(t, x) & t \geq 0, x \in \mathbb{R} \\ u(0, x) = f(x) & x \in \mathbb{R} \end{cases}$$

The solution to the stochastic heat equation is given through the following integral equation:

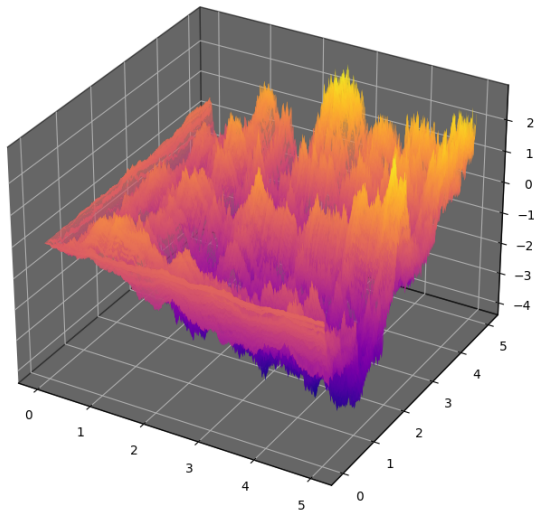
$$u(t, x) = \int_{\mathbb{R}} G(t, x - y) f(y) dy + \lambda \int_0^t \int_{\mathbb{R}} G(t - s, x - y) u(s, y) W(ds, dy),$$

where

$$G(t, x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{|x|^2}{2t}\right)$$

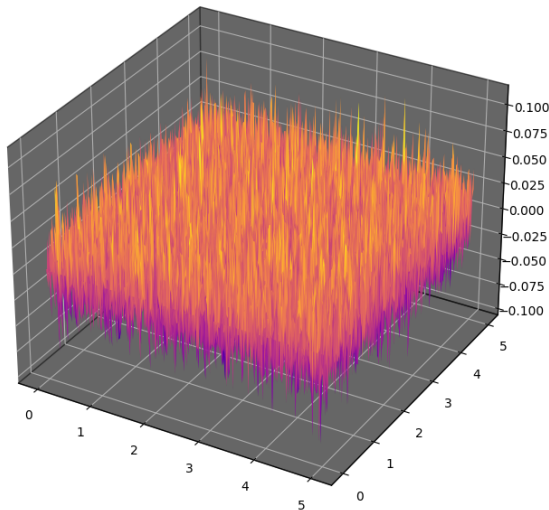
Sample Path of a Brownian Sheet, $W(t, x)$

Brownian Sheet



Approximation of a Space-Time White Noise Differential

Space-time white noise: $\dot{W}(t, x) \approx \int_{t-\delta t}^t \int_{x-\delta x}^x dW(s, y)$



Random vs Deterministic

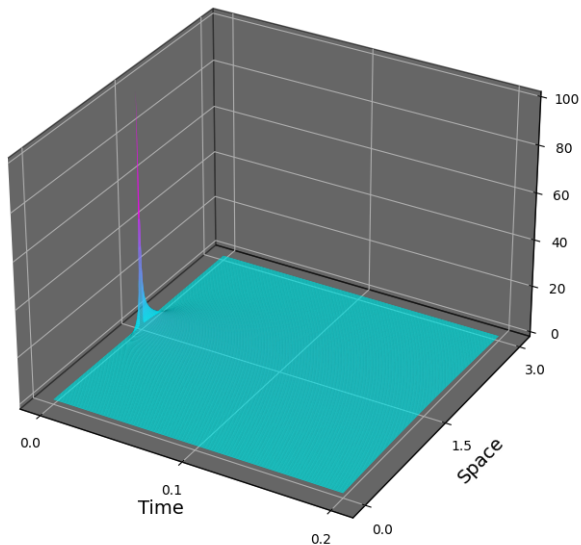
- The solution to the stochastic heat equation is a stochastic process.

$$u(t, x)(\omega_1), \quad u(t, x)(\omega_2), \quad \cdots, \quad u(t, x)(\omega_n).$$

- The random solution only exists implicitly through the integral equation.
- $\frac{\partial}{\partial x} u(t, x)$ and $\frac{\partial}{\partial t} u(t, x)$ does not exist in the random case with space-time white noise.
- Interesting 'blow up' happens in the random case. This actually has real-world application.

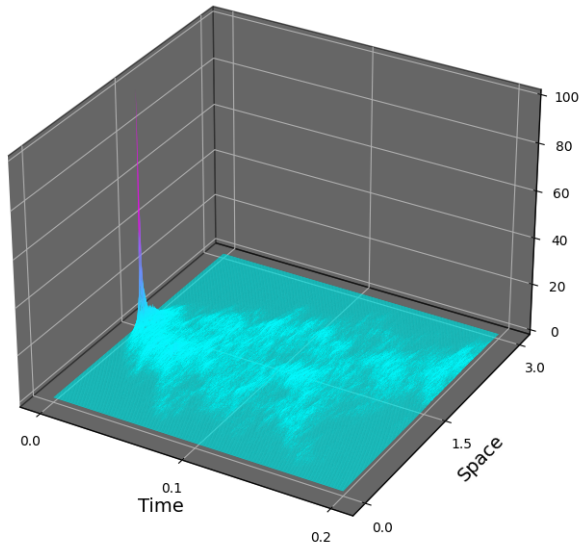
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 0.0u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



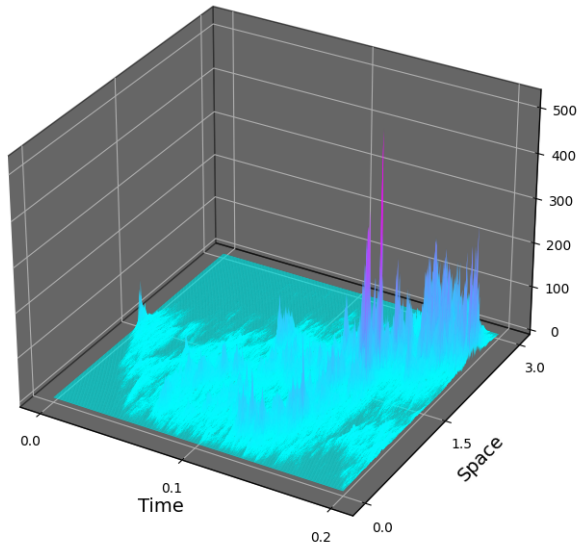
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 1.0u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



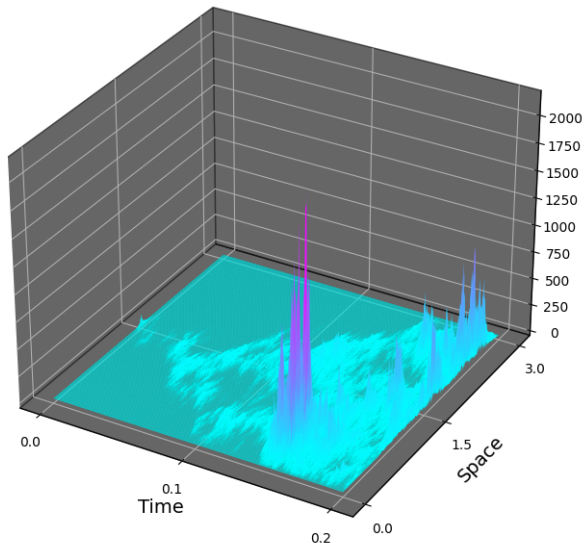
$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 2.05u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



$$u_t(t, x) - \frac{1}{2}u_{xx}(t, x) = 2.15u(t, x)\dot{W}(t, x)$$

$$u(x, 0) \approx \delta_0(x)$$



Growth of the Paths and Moments

- Can we accurately describe the creation of these large peaks? In other words, how can we describe the sample paths of the solution:

$$t \mapsto u(t, x)(\omega) \quad \text{and} \quad x \mapsto u(t, x)(\omega)$$

- This is a hard question and a current open problem is calculate the following limit

$$\lim_{t \rightarrow \infty} \frac{\log |u(t, x)|}{t^K} = \begin{cases} 0 & t \mapsto u(t, x) \text{ grows slower than } e^{t^K} \\ C & t \mapsto u(t, x) \text{ grows the same as } e^{C t^K} . \\ \infty & t \mapsto u(t, x) \text{ grows faster than } e^{t^K} \end{cases}$$

- The following has been calculated for the SHE and SWE (and also other SPDEs which we will see below)

$$\lim_{t \rightarrow \infty} \frac{\log [\mathbb{E}(|u(t, x)|^p)]}{t^K}, \quad p \geq 2.$$

The Stochastic Heat and Wave Equations

$$\begin{cases} (\partial_t^b - \Delta_x) u(t, x) = u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b = 1 \quad (\text{SHE}) \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b = 2 \quad (\text{SWE}) \end{cases}$$

- $\dot{W}$ is a time-independent Gaussian noise defined on $\phi, \psi \in C_0^\infty(\mathbb{R}^d)$:

$$\mathbb{E}(W(\phi)W(\psi)) = \int_{\mathbb{R}^{2d}} dx dy \phi(x) |x - y|^{-\alpha} \psi(y)$$

Known Results

- For the SHE, i.e., $b = 1$
[X. Chen '17]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{2-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{2}{2-\alpha}} (2-\alpha) \left(\frac{2\mathcal{M}}{4-\alpha} \right)^{\frac{4-\alpha}{2-\alpha}}$$

- For the SWE, i.e., $b = 2$
[Balan, L. Chen, and X. Chen '21]

$$\lim_{t \rightarrow \infty} t^{-\frac{4-\alpha}{3-\alpha}} \log \mathbb{E}|u(t, x)|^p = p(p-1)^{\frac{1}{3-\alpha}} \left(\frac{1}{2} \right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha} \right)^{\frac{4-\alpha}{3-\alpha}}$$

The Interpolated Stochastic Heat and Wave Equation

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [\sqrt{\theta} u(t, x) \dot{W}(x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (\text{ISHWE})$$

- ∂_t^b is the *Caputo* fractional derivative of order $b > 0$
- $(-\Delta)^{a/2}$ is the fractional Laplacian of order $a \in (0, 2]$
- I_t^r is the Riemann-Liouville fractional integral of order $r \geq 0$
- $\dot{W}$ is a time-independent Gaussian noise defined on $\phi, \psi \in C_0^\infty(\mathbb{R}^d)$:

$$J(\phi, \psi) := \mathbb{E}(W(\phi)W(\psi)) = \int_{\mathbb{R}^d} dx \, \phi(x) |x - y|^{-\alpha} \psi(y)$$

Fractional Differential Operators

- We recall a property of the standard Laplacian, $\Delta = \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_d^2}$:

$$(-\Delta)f(x) = \left((|\cdot|^2)\hat{f}(\cdot) \right)^\vee(x).$$

This gives motivation for the fractional Laplacian:

$$(-\Delta)^{\alpha/2}f(x) := \left((|\cdot|^\alpha)\hat{f}(\cdot) \right)^\vee(x),$$

- The Caputo derivative is defined as

$$\partial_t^b f(t) := \begin{cases} \frac{1}{\Gamma(m-b)} \int_0^t d\tau \frac{f^{(m)}(\tau)}{(t-\tau)^{b+1-m}} & \text{if } m-1 < b < m, \\ \frac{d^m}{dt^m} f(t) & \text{if } b = m, \end{cases}$$

Mild Solution

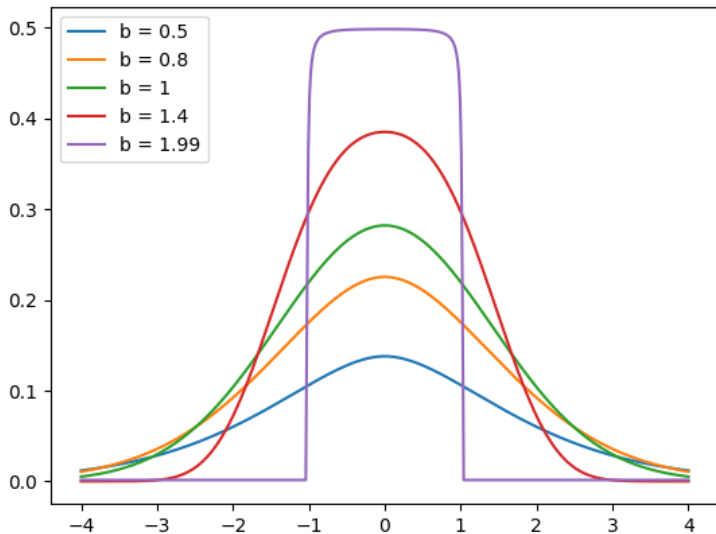
- The solution to the ISHWE is any processes that satisfies the following stochastic integral equation:

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds, \quad (0.1)$$

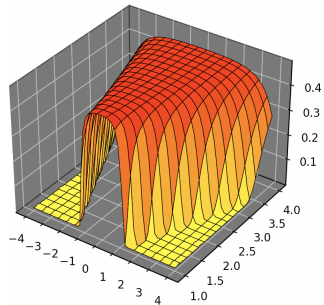
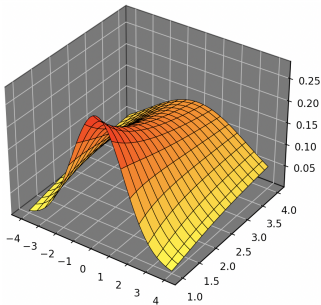
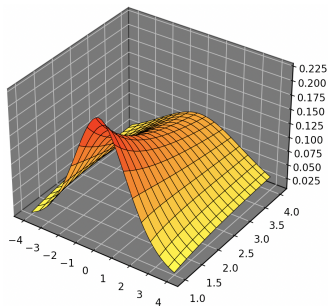
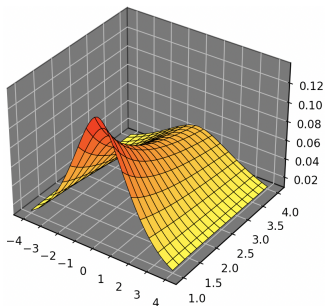
- The kernal G is defined through the Fox-H function:

$$\begin{aligned} & G_{a,b,r,\nu,d}(t, x) \\ &= \pi^{-d/2} |x|^{-d} t^{b+r-1} H_{2,3}^{2,1} \left(\frac{|x|^a}{2^{a-1} \nu t^b} \left| \begin{array}{l} (1, 1), (b+r, b) \\ (d/2, a/2), (1, 1), (1, a/2) \end{array} \right. \right). \end{aligned}$$

$G(1, x)$ with $a = 2, \gamma = 0, \nu = 2, d = 1$ and $.5 \leq b < 2$



$G(t, x)$ with $b = .5, .8, 1, 1.9$ and $1 \leq t \leq 4$



Global Solution: $(\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right]$

Definition

$u(t, x)$ is a *global solution* to the (ISHWE) if $\|u(t, x)\|_p < \infty$ for any $0 < t < T$ and $x \in \mathbb{R}^d$ where $T > 0$ is arbitrary.

Theorem (Chen-E. '21+)

A global solution exists provided G is nonnegative and

$$0 < \alpha < \min \left(\frac{a}{b} [2(b+r) - 1], 2a, d \right)$$

Local Solution: $(\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [\sqrt{\theta} u(t, x) \dot{W}(x)]$

Definition

$u(t, x)$ is a *local solution* to the (ISHWE) if there exists $0 < T_a \leq T_b < \infty$ such that $\|u(t, x)\|_2 < \infty$ when $0 < t < T_a$ and $\|u(t, x)\|_2$ D.N.E. (does not exist) for $t > T_b$.

Theorem (Chen-E. '21+)

A local solution exists provided G is nonnegative and if

$$r \in [0, 1/2] \quad \text{and} \quad 0 < \alpha = \frac{a}{b}[2(b+r) - 1] \leq d.$$

In this case, a unique local $L^2(\Omega)$ solution exists for $t \in (0, T_2)$ where

$$T_p := \frac{\nu^{\alpha/a}}{2\theta \mathcal{M}_a^{(2a-\alpha)/a}}$$

and the solution does not exist for $t > T_2$.

Special Form of the Solution

- The time independent noise and the constant initial conditions allows us to prove the following explicit representation of the solution:

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)), \quad (t, x) \in (0, T) \times \mathbb{R}^d$$

$$f_n(x_1, \dots, x_n; x, t) = \int_0^t \int_0^{t_n} \cdots \int_0^{t_2} \prod_{k=1}^n G(t_{k+1} - t_k, x_{k+1} - x_k) dt_1 \cdots dt_n$$

- Moreover, we can show that the convergence of the above series is equivalent to showing the following:

$$\|u(t, x)\|_2^2 = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 < \infty$$

Exact Moment Lyapunov Exponent

(Recall $\beta := \frac{2(b+r) - \frac{b\alpha}{a}}{2(b+r) - \frac{b\alpha}{a} - 1}$)

Corollary (Chen-E. '21+)

For $p \geq 2$ fixed, we have that

$$\begin{aligned} & \lim_{t \rightarrow \infty} t^{-\beta} \log \mathbb{E}(|u(t, x)|^p) \\ &= p(p-1)^{\frac{1}{2(b+r) - \frac{b\alpha}{a} - 1}} \left(\frac{1}{2}\right) \left(\frac{2a}{2a(b+r) - b\alpha}\right)^{\beta} \\ & \quad \times \left(\theta \nu^{-\alpha/a} \mathcal{M}_a^{\frac{2a-\alpha}{a}}\right)^{\frac{a}{2a(b+r) - b\alpha - a}} \left(2(b+r) - \frac{b\alpha}{a} - 1\right). \end{aligned}$$

From this we can deduce that $t \mapsto \mathbb{E}(|u(t, x)|^p)$ will grow like $\exp(K_1 t^\beta)$, in other words, the moments will grow exponentially in time, and we obtain the exact expression for the constant K_1 .

The Case of a time-dependent Noise and super-linear diffusion term

$$\begin{cases} (\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [\sigma(u(t, x)) \dot{W}(t, x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & b \in (0, 1] \\ u(0, \cdot) = 1, \quad \partial_t u(0, \cdot) = 0 & b \in (1, 2) \end{cases} \quad (0.2)$$

- The diffusion coefficient has super-linear growth:

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta \quad \text{as } |x| \rightarrow \infty.$$

- The noise is white in time and either white or colored in space, ie for $\phi, \psi \in C_0^\infty(\mathbb{R}_+ \times \mathbb{R}^d)$

$$\mathbb{E}(W(\phi)W(\psi)) = \begin{cases} \int_0^\infty ds \int_{\mathbb{R}^d} dx \phi(s, x) \psi(s, x) \\ \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x) |x - y|^{-\alpha} \psi(s, y). \end{cases}$$

Results when $|\sigma(x)| \leq |\sigma(0)| + L(\sigma)|x|$

- A global solution exists when $\rho := 2b + 2r - 1 - bd/a > 0$ and $d < 2a$.
- When a global solution exists, then there exists a universal constant $K := K(\alpha, \beta, \gamma, d) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2}\right],$$

Then for any $T > 0$,

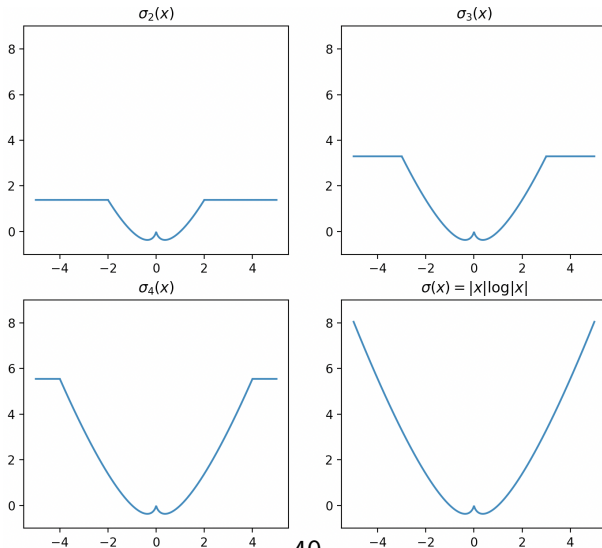
$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)}\right]^p, \quad t \in [0, T]. \quad (0.3)$$

- The solution $u(t, x)$ is continuous in t and in x .

Existence of the Solution for Super-linear σ Term

We consider a truncated version of the equation where σ replaced by σ_N :

$$\sigma_N(x) = \sigma(x)\mathbf{1}_{\{|x| \leq N\}} + \sigma(N)\mathbf{1}_{\{x > N\}} + \sigma(-N)\mathbf{1}_{\{x \leq -N\}}.$$



Sketch

- We see that σ_N is Lipschitz:

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|,$$

so there exists a unique solution that is continuous, which we denote as $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$.

- When

$$2b + 2r - 1 - bd/a > 0 \quad \text{and} \quad d < 2a,$$

$u_N \rightarrow u$ which satisfies

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)| < \infty, \text{ a.s.}$$

- The convergence is not $\mathbb{E}(|u_N - u|^2) \rightarrow 0$, which is the usual convergence that is proven. What we have is

$$\lim_{N \rightarrow \infty} \mathbb{P}(|u_N - u|) \rightarrow 0.$$

Thank you!